

Ann Song

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EDUCATION

Yale University

BS in Computer Science and Mathematics | GPA: 3.93/4.0

TA for CPSC 2020 Mathematics for Computer Science, Freshmen Pre-Orientation Leader, Summer 2026 Tour Guide

New Haven, CT

June 2029

WORK EXPERIENCE

Medtronic (Surgical Robotics Operating Unit)

Software Engineer Intern

North Haven, CT

June 2026 – Present

- Redesigned a legacy Perl log-parsing pipeline for the Signia powered surgical stapling platform into a modular Python implementation, 3× faster on a full device log and expanding coverage to 12 record types across 4 firmware log versions.
- Interviewed Release Product Analysis engineers to scope their diagnostic workflow, then built an analysis UI over the parser that collapsed returned-device triage from a manual multi-tool process into a single view, reducing time-to-diagnosis by 60%.
- Developed automated motor telemetry visualization from binary .rdf captures, segmenting each run into clamp, fire, and cut phases so engineers can localize a device anomaly to one stage of the firing cycle.
- Interviewed 14 Release Product Engineers and R&D stakeholders to identify diagnostic questions existing logs could not answer; worked with the Micrium RTOS on the motor control path to log scheduling jitter, task utilization, and reset and fault causes persisted across reset, enabling engineers to distinguish mechanically anomalous fires from software-induced ones.

AI Engineer (Contract)

Oct 2025 – May 2026

Software Engineer Intern

June 2025 – Aug 2025

- Scoped the first tool-augmented RAG agent for Software Failure Modes and Effects Analysis (SFMEA) through requirements interviews from sales reps with systems engineering and 12 quality engineers as design partners; built and shipped the production system to 800+ surgical stapler sales reps, cutting engineer escalations 65% and response turnaround from days to minutes.
- Designed the retrieval layer over 10,000+ pages of internal technical documentation, including semantic chunking and a document classifier separating specification from device architecture content, deployed as Python microservices in CI/CD architecture.
- Built an evaluation system to score six LLM models on cost, latency, and hallucination rate against surgeon question sets elicited from field sales representatives, which cut original inference costs by 60% with the same response accuracy.

Yale Inquisitive Robotics Lab

Undergraduate Researcher in Physical AI

New Haven, CT

Sept 2025 – Present

- Developing introspective failure prediction for Vision-Language-Action (VLA) policies by analyzing internal transformer activations and multimodal rollout dynamics from π_{FAST} and $\pi_{0.5}$ across 3,000+ LIBERO benchmark rollouts; second-author manuscript in submission to ICLR 2027.
- Evaluated 40+ diagnostic features spanning saliency concentration, temporal drift, value geometry, policy planning, and cross-modal agreement to predict rollout failure before visible task divergence, enabling earlier robot recovery.
- Built sequential failure prediction pipelines using GRU temporal models over 512-dimensional transformer representations, trained across 20 randomized seeds with rollout-level train/validation/test splits.

PROJECTS

Autonomous Tomato Harvesting Robot (2-Person Team)

ROS 2, Python, OpenCV, YOLOv11, Gazebo, React, TypeScript, Raspberry Pi

New Haven, CT

June 2026 – Present

- Built an autonomous cherry tomato harvesting system in ROS 2 on a custom 4-DOF arm with a vacuum end effector, integrating perception, stereo depth, kinematics, and closed-loop control; validated by picking real cherry tomatoes in field testing.
- Developed a feedback-driven motion controller that converts Cartesian waypoints into coordinated joint trajectories, synchronizing per-joint speeds so the end effector follows a straight approach path; then implemented a closed-form 4-DOF inverse-kinematics solver and a staged pregrasp-contact-retreat approach planner that generate reachable trajectories from the robot URDF.
- Developed stereo depth pipeline (monocular/stereo calibration, rectification, semi-global block matching) with two Raspberry Pi Cameras with an eye-to-hand transform to localize each tomato in 3D relative to the arm.
- Deployed a YOLOv11s detector with persistent multi-target tracking and HSV color-coverage ripeness fusion, tuned for precision over recall to avoid unripe fruit.
- Built a browser-based control dashboard (React/TypeScript, roslib over rosbridge) for arm teleoperation and tomato target selection; set up a Gazebo simulation from the robot's URDF and ran the full controller in it, using ground-truth tomato poses to isolate whether reach failures came from stereo-depth estimation or the IK/motion planner.

SKILLS AND INTERESTS

Languages: Python [NumPy, SciPy, PyTorch, Pandas, OpenCV], C++, C, Java, JS [ReactJS, TypeScript], MATLAB, Bash

Robotics: ROS 2, URDF, TF2, RViz, Gazebo, Foxglove, stereo calibration, SGBM, forward/inverse kinematics, YOLO

Tools: Git, Docker, Linux, AWS, JAX/Flax, Slurm/HPC, Claude Code, Cursor

Interests: Yale Red Lantern (Traditional Chinese Dance), Yale Figure Skating Club, Weightlifting